Decision Tree

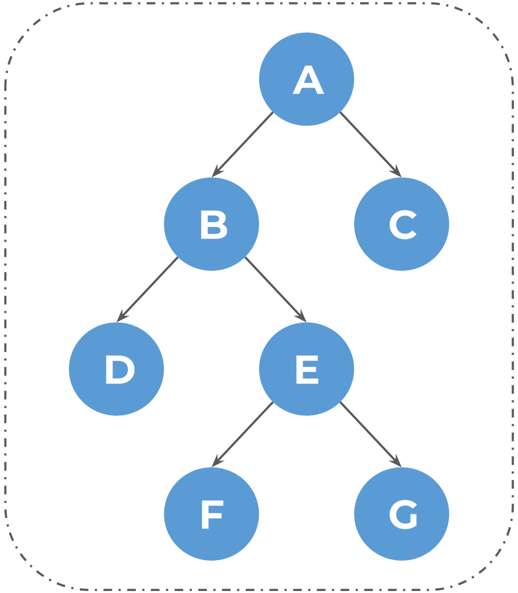
Agenda

- Introduction to Decision Tree
- Pruning Techniques: Pre-Pruning vs Post-Pruning
- Impurity Measures for Decision Trees (Classification)

Let's begin the discussion by answering a few questions

Classification Methods Quiz

Which of the following nodes are leaf nodes?



A — A, B, E

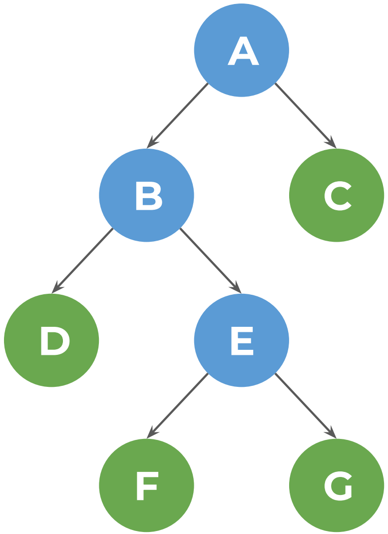
B — F, G

C — C, D, F, G

D — A, B, C

Classification Methods Quiz

Which of the following nodes are leaf nodes?



A. A, B, E

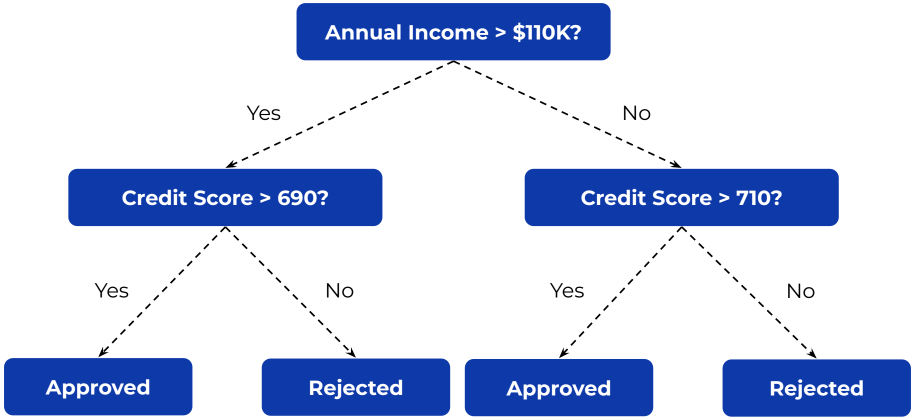
B. F, G

C. C, D, F, G

D. A, B, C

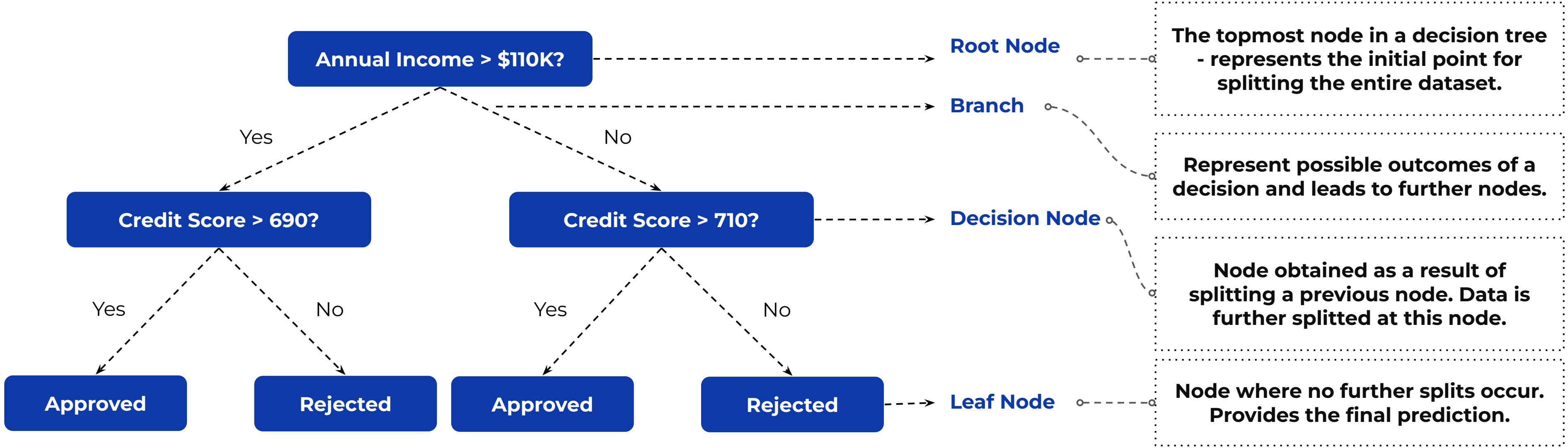
Introduction to Decision Tree

A **decision tree is** a popular and powerful supervised learning algorithm that works with both categorical and continuous variables by visually representing all possible decisions based on given conditions, where the training data is repeatedly split into smaller groups according to defined rules to achieve better prediction accuracy.



Unlike linear regression, which mainly captures straight-line (linear) relationships between features and the target, decision trees can capture more complex patterns.

Decision Tree Structure



At the **leaf node**, find the **proportion of each class**

The **class** with the **higher proportion** is the **prediction**

Decision Tree Quiz

You are building a Decision Tree model for fraud detection with a large dataset and limited computational resources. To ensure faster training and prevent overfitting, which pruning strategy should you choose?

A. Post-Pruning

B. Pre-Pruning

C. Both Pre-Pruning and Post-Pruning

D. No pruning is required

Decision Tree Quiz

Great Learning
POWER AHEAD

You are building a Decision Tree model for fraud detection with a large dataset and limited computational resources. To ensure faster training and prevent overfitting, which pruning strategy should you choose?

A. Post-Pruning

B. Pre-Pruning

C. Both Pre-Pruning and Post-Pruning

D. No pruning is required

Pre-Pruning vs Post-Pruning

Decision trees can become too large and overfit the training data, so pruning simplifies the tree by removing unnecessary branches and converting some branch nodes into leaf nodes.

Pre-Pruning	Post-Pruning
Prevents the decision tree from **growing fully** by introducing a **stopping criterion**	Allows the **decision tree to grow fully** and then removes **unnecessary branches and/or leaves**
Prevents the tree from **capturing noise and overfitting** on the training data	The **Nodes and Subtrees** are **replaced** with **leaves** to **reduce complexity**
Faster and computationally efficient as the entire tree is not built	**Slower and computationally expensive** as it requires building the entire tree first before trimming unnecessary branches

Decision Tree Quiz

Which of the following parameters is not used for pre-pruning a Decision Tree?

A `max_depth`

B `min_samples_leaf`

C `ccp_alpha`

D `min_samples_split`

Decision Tree Quiz

Which of the following parameters is not used for pre-pruning a Decision Tree?

A `max_depth`

B `min_samples_leaf`

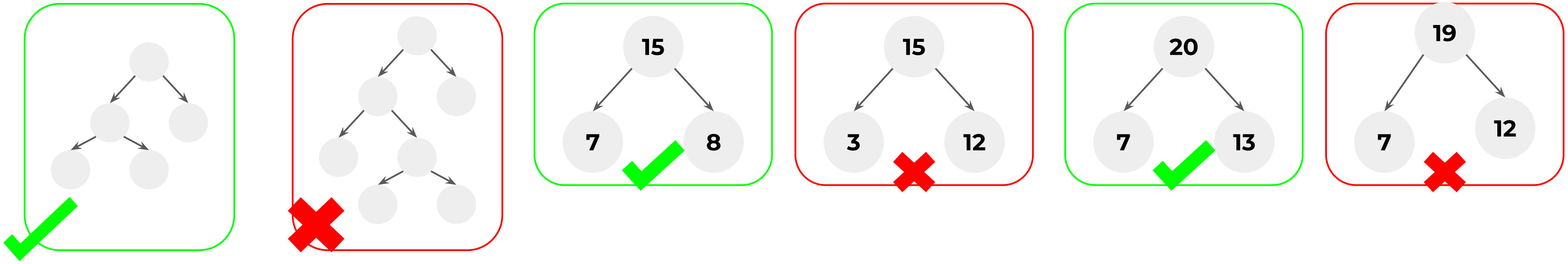
C `ccp_alpha`

D `min_samples_split`

Pre-Pruning a Decision Tree

Hyperparameters that help pre-prune a decision tree

Max Depth	Min Samples for Leaf	Min Samples to Split
Sets the maximum depth to which a tree can grow	The minimum number of samples to be present in every leaf node	The minimum number of samples to be present in a node for it to be eligible for a split
max_depth = 2	min_samples_leaf = 5	min_samples _split = 20



Post-Pruning a Decision Tree

Hyperparameter that help post-prune a decision tree

Cost Complexity Pruning

ccp_alpha

Balances tree complexity with classification accuracy by penalizing the number of nodes

Decision Tree Quiz

Which of the following attribute-threshold combinations should be chosen for the first split in a decision tree based on Gini Impurity reduction?

- **Annual Income > $110K | Gini before split = 0.43, after split = 0.37**
- **Credit Score > 690 | Gini before split = 0.45, after split = 0.4**
- **Annual Income > $70K | Gini before split = 0.45, after split = 0.4**
- **Credit Score > 700 | Gini before split = 0.44, after split = 0.41**

A. Credit Score > 690

B. Annual Income > $ 70k

C. Annual Income > $ 110k

D. Credit Score > 700

Decision Tree Quiz

Which of the following attribute-threshold combinations should be chosen for the first split in a decision tree based on Gini Impurity reduction?

- **Annual Income > $110K | Gini before split = 0.43, after split = 0.37**
- **Credit Score > 690 | Gini before split = 0.45, after split = 0.4**
- **Annual Income > $70K | Gini before split = 0.45, after split = 0.4**
- **Credit Score > 700 | Gini before split = 0.44, after split = 0.41**

A. Credit Score > 690

B. Annual Income > $ 70k

C. Annual Income > $ 110k

D. Credit Score > 700

Impurity Measures

Gini Impurity

Measures the probability of misclassifying a randomly selected data point if we label it randomly

Higher the Gini Impurity, lower the homogeneity - worse the split

Ranges between 0 (pure) and 0.5 (impure)

$$Gini\ Impurity = 1 - \sum_{i=1}^{k} p_i^2$$

Entropy

Measures how mixed up or organized data points are in a certain group.

Higher the Entropy, lower the homogeneity - worse the split

Ranges between 0 (pure) and 1 (impure)

$$Entropy = -\sum_{i=1}^{k} p_i \log_2(p_i)$$

p_i = proportion (probability) of class i in the node; k = total number of classes

Decision Tree Quiz

A company is developing a medical diagnostic tool to detect a rare disease. Which metric should they prioritise when evaluating their model?

A. Accuracy - to ensure the model is generally correct

B. Precision - to reduce the number of healthy people flagged as sick

C. Recall - to identify as many actual patients as possible

D. F1-score - to balance precision and recall

Decision Tree Quiz

A company is developing a medical diagnostic tool to detect a rare disease. Which metric should they prioritise when evaluating their model?

A. Accuracy - to ensure the model is generally correct

B. Precision - to reduce the number of healthy people flagged as sick

C. Recall - to identify as many actual patients as possible

D. F1-score - to balance precision and recall

Confusion Matrix

A table that helps you visualize the performance of a classification model

Especially useful when you have a binary classification problem

We have four possible outcomes from a classification model:

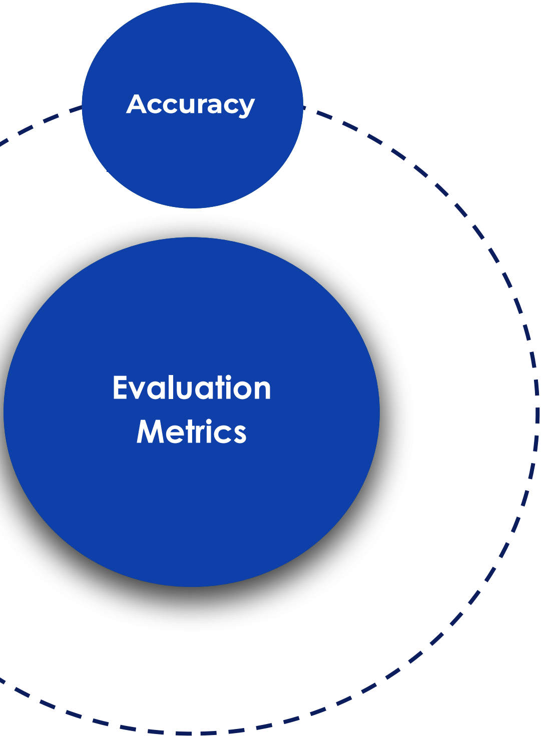
	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Understanding Model Predictions

Consider the previous problem of detection of a rare disease

True Negative (TN)	False Positive (FP)	True Positive (TP)	False Negative (FN)
Actual Negative Predicted Negative	Actual Negative Predicted Positive	Actual Positive Predicted Positive	Actual Positive Predicted Negative
Case 1	**Case 2**	**Case 3**	**Case 4**
The patient did not have the disease and the model also predicted that the patient did not have the disease	The patient did not have the disease but the model predicted that the patient had the disease	The patient had the disease and the model also predicted that the patient had the disease	The patient had the disease but the model predicted that the patient did not have the disease

Classification Metrics

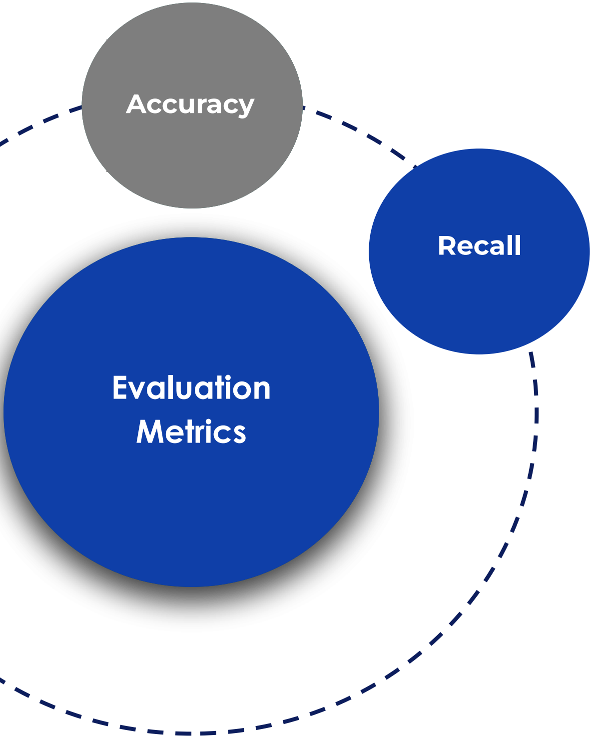


- Most intuitive and widely used metric

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- Represents the **proportion of correct predictions** out of the total number of predictions
- Ranges from 0 to 1

Classification Metrics



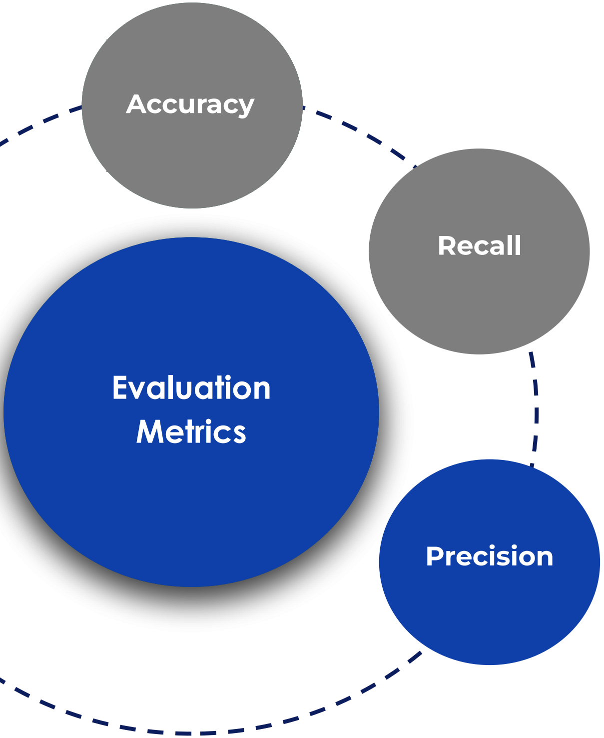
- The ratio of true positives to total actual positives

$$Recall = \frac{TP}{TP + FN}$$

- Tells us out of all the ‘positive’ instances, how many did the model remember correctly
- Ranges from 0 to 1

Recall ↑ **False Negatives** ↓

Classification Metrics



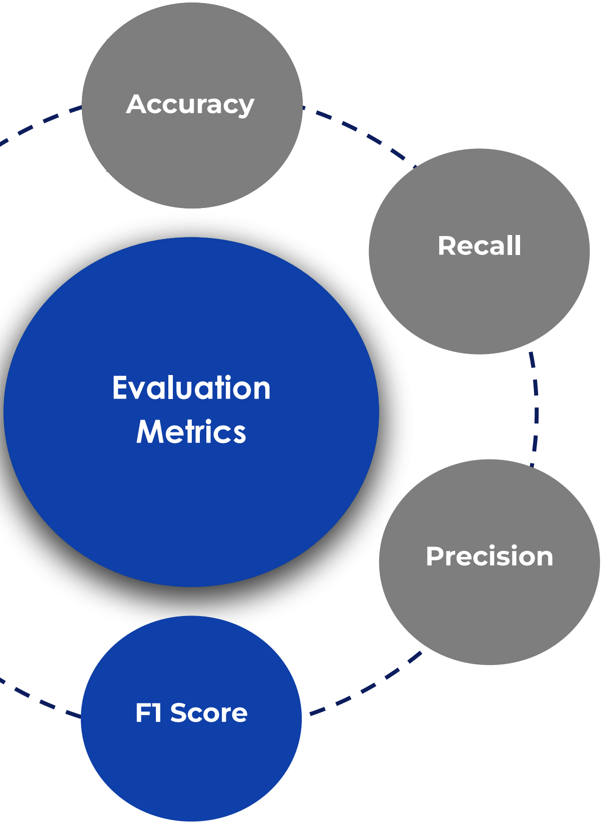
- The ratio of true positives to total predicted positives

$$Precision = \frac{TP}{TP + FP}$$

- Tells us out of all the predicted 'positive' instances, how many are actually 'positive'
- Ranges from 0 to 1

Precision ↑ **False Positives** ↓

Classification Metrics



- Maintains a balance between False Positives and False Negatives

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- Ranges from 0 to 1

F1 Score ↑ **False Positives** ↓ **False Negatives** ↓

Power Ahead !

Appendix: Classification vs Regression Trees

Classification Trees	Regression Trees
Predict a categorical value (class label)	Predict a numerical value
Uses Impurity measures like Gini Impurity and Entropy for splitting	Uses measures like Mean Squared Error (MSE) for splitting
Focuses on reducing impurity with each split	Focuses on reducing variance with each split